

Plants, Part 2: Photosynthesis

Answer sheet · 30 marks total

Section A — Where photosynthesis happens

A1. Chloroplast (1).

A2. Chlorophyll (1).

A3. Being near the top of the leaf means they receive more light (1); having many chloroplasts means more chlorophyll is available to absorb it, increasing the rate of photosynthesis (1).

Section B — The photosynthesis equation

B1. Carbon dioxide + water → glucose + oxygen (2, allow 1 if one side is wrong/missing).

B2. Carbon dioxide diffuses in through the stomata (1); water is delivered by the xylem (1).

B3. Endothermic reactions take in energy from the surroundings (1); photosynthesis takes in light energy to drive the reaction (1).

B4. Any three: used for respiration (1); converted to starch for storage (1); used to make cellulose for cell walls (1); used to make plant proteins (1); converted to sucrose for transport (1).

Section C — Factors affecting the rate

C1. Light intensity (1); carbon dioxide concentration (1); temperature (1).

C2. CO₂ is a raw material/reactant for photosynthesis (1); more of it available means the reaction can proceed faster (1).

C3. High temperatures denature the enzymes needed for the reaction (1); denatured enzymes can no longer catalyse the reaction efficiently, so the rate falls (1).

C4. The limiting factor (1).

C5. Another factor (e.g. CO₂ or temperature) has become the limiting factor (1); increasing light intensity alone cannot raise the rate further until that other factor increases too (1).

Section D — Translocation, nutrition & disease

D1. The movement of dissolved sugars (1) around the plant through the phloem (1).

D2. Phloem (1).

D3. Magnesium (1); magnesium is needed to make chlorophyll, so a shortage reduces the green pigment in the leaves (1).

D4. Nitrate (1); nitrate is needed to make proteins, which are needed for cell growth (1).

D5. Any three: stunted growth (1); spots on leaves (1); patches of decay/rot (1); growths/galls (1); discolouration (1); wilting (1).

Marking note: accept any scientifically correct alternative answer with the same reasoning, even if different from the example shown — award marks for a correctly linked point, not for matching these exact words.